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AI INNOVATION HACKATHON 2026

Track 3: Explainable AI (XAI) | Stage 1: Idea Screening

ChurnLens

Explainable, Actionable and Trust-Scored Customer Churn Prediction

Challenge 08: Explain My Customer Churn Prediction

TEAM VISIONARIES

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Problem Statement

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THE PROBLEM

Businesses use AI to predict whether a customer is likely to leave, but a churn score alone does not explain what contributed to the prediction. Retention teams cannot tell why a customer is at risk, whether the explanation can be trusted, or what action could prevent the loss. Opaque scores lead to poorly targeted retention offers, wasted budget and low confidence in AI-driven decisions.

1. Black-box scores

A probability such as 0.78 gives no reasons. Teams cannot justify why a specific customer was flagged.

2. No measure of trust

Even when explanations exist, nobody checks whether they are stable or consistent across similar customers.

3. No path to action

Knowing the drivers of churn is not enough. Teams need to know what to change to reduce the risk.

Why it matters

Acquiring a new customer typically costs far more than retaining an existing one. An explainable and trustworthy churn model lets retention teams focus on the right customers with the right intervention.

FROM PREDICTION TO EXPLANATION TO ACTION

Proposed Solution: ChurnLens

ChurnLens is a churn prediction tool where every prediction is explained, checked for reliability and converted into practical retention suggestions.

1 PREDICT

Enter customer details and get a churn probability.

2 EXPLAIN

Local SHAP explanation shows the top factors pushing the risk up or down.

3 SUMMARIZE

Factors are turned into a short plain-English sentence for managers.

4 ACT

What-if engine tests changes to actionable features and shows the risk reduction.

5 TRUST

A trust score shows how consistent and reliable the explanation is.

On-screen demonstration flow

Customer details → **Churn prediction** → **Top influencing factors** → **Explanation + trust score + what-if**

- Interface: Streamlit web app. Output is shown as a risk alert with reasons, never as a final verdict.

WHAT MAKES CHURNLENS DIFFERENT

Innovation and Novelty

1. Actionable What-if Retention Engine

Goes beyond why a customer may churn to how the risk could be reduced.

- Changes only actionable features: contract type, tech support, online security, payment method
- Re-predicts and ranks changes by risk reduction
- Example: month-to-month to one-year contract lowers predicted risk

2. Explanation Trust Score

The challenge asks for explanation consistency. We measure it instead of assuming it.

- SHAP vs LIME top-5 feature agreement
- Explanation similarity across nearest-neighbour customers
- Shown as a confidence indicator next to every explanation

3. Plain-Language Output + Fairness Audit

Makes the model usable by non-technical teams and checks it is not biased.

- SHAP factors converted into a readable sentence for retention managers
- Group-wise metric comparison across gender and senior-citizen groups
- Sensitive attributes are never used as recommended actions

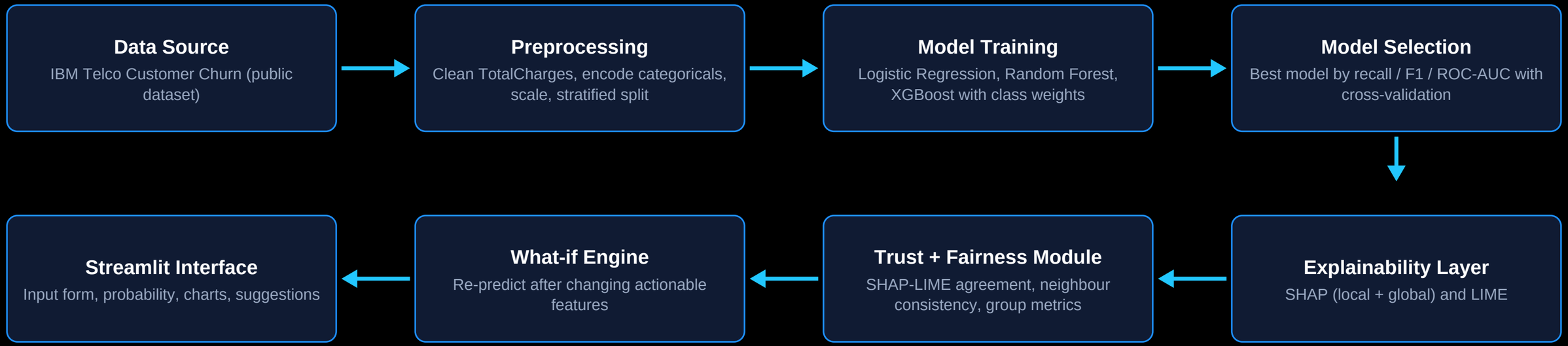
OUR EDGE

Most XAI projects stop at a SHAP plot. ChurnLens connects the full chain: prediction, explanation, reliability check, fairness check and recommended action, so the explanation becomes a tool for decisions rather than just a chart.

Technical feasibility: built entirely on open-source Python libraries and a public dataset.

END-TO-END PIPELINE

Solution Architecture



WHAT THE USER SEES

- Churn probability with a clear risk level (low / medium / high).
- Top factors increasing or decreasing risk, with a one-sentence plain-language summary.
- Trust score for the explanation and a ranked list of retention actions with estimated risk reduction.

TECHNIQUES, ALGORITHMS AND DATA

AI/ML Methodology and Approach

Dataset and Preprocessing

- IBM Telco Customer Churn dataset (about 7,000 customers, 20+ attributes, roughly one quarter churners)
- Convert TotalCharges to numeric and handle missing values
- One-hot / ordinal encoding for categorical features
- Scale numeric features where required
- Stratified train/test split to preserve churn ratio
- Public, permitted data only

Classification Models

- Logistic Regression: interpretable baseline
- Decision Tree / Random Forest: non-linear patterns
- XGBoost / Gradient Boosting: expected top performer
- Class imbalance handled with class weights and decision-threshold tuning
- Hyperparameter tuning with cross-validated search
- Final model chosen on recall/F1 balance and explanation stability

Explainability and Trust

- SHAP for local (per customer) and global feature importance
- LIME as an independent second explanation method
- Trust score = SHAP-LIME top-k overlap plus similarity of explanations among nearest-neighbour customers
- What-if analysis on actionable features only
- Fairness audit by gender and senior-citizen groups

OPEN-SOURCE, REPRODUCIBLE AND FEASIBLE

Technology Stack

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LANGUAGE

Python 3

DATA HANDLING

pandas, NumPy

MODELLING

scikit-learn, XGBoost

IMBALANCE

class weights, threshold tuning

EXPLAINABILITY

SHAP, LIME

VISUALISATION

matplotlib, SHAP plots

INTERFACE

Streamlit

ENVIRONMENT

Jupyter Notebook, Google Colab

VERSION CONTROL

Git and GitHub

Everything runs on free tools and a free public dataset, so the full pipeline can be reproduced by anyone.

HOW WE WILL PROVE IT WORKS

Validation Plan and Evaluation Metrics

Metric	What it measures	Why we use it
Precision	Share of predicted churners who really churn	Avoids wasting retention offers
Recall	Share of real churners we catch	Avoids missing at-risk customers
F1-score	Balance of precision and recall	Single score under class imbalance
ROC-AUC	Ranking quality across all thresholds	Threshold-independent performance
PR-AUC	Precision-recall trade-off	Better view for imbalanced churn data
Explanation consistency	SHAP-LIME overlap, neighbour similarity	Tests if explanations can be trusted
Fairness gap	Metric difference across groups	Checks the model treats groups similarly

VALIDATION STEPS

- Stratified train/test split and 5-fold cross-validation
- Compare all models; select on recall/F1 plus stability
- Threshold analysis to balance missed churners vs false alarms
- Repeat explanation checks across random seeds and customer groups

BUILT TO BE USED SAFELY

Responsible AI and Ethical Considerations

Alerts, not verdicts

Predictions are shown as probabilistic risk indicators. Explanations describe model reasoning, not guaranteed causes of customer behaviour.

Public and permitted data

Only the public IBM Telco dataset is used. No personal or confidential customer data is collected or stored.

Fairness audit

Model performance is compared across gender and senior-citizen groups, and any gap is reported openly.

Safe recommendations

What-if suggestions only change business-controlled features such as contract, support and payment options, never personal attributes.

Transparency of limits

Model limits, class imbalance and the trust score are shown in the interface so users do not over-rely on the output.

Human in the loop

The tool supports retention teams in making decisions. Final action stays with a human decision-maker.

VALUE AND NEXT STEPS

Real-World Impact and Roadmap

Real-world impact

- Retention teams can prioritise the customers who matter most, with clear reasons
- Targeted interventions reduce wasted discounts and campaign cost
- Trust scores help managers decide how far to rely on each explanation
- Fairness checks support responsible deployment in telecom, banking, SaaS and subscription businesses
- Same framework can be reused for other tabular prediction problems

Round 2 build plan

1. **Phase 1**
   Data cleaning, baseline models, evaluation
2. **Phase 2**
   SHAP and LIME explanations, trust score
3. **Phase 3**
   What-if engine and fairness audit
4. **Phase 4**
   Streamlit interface, demo flow and documentation

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Thank You

ChurnLens: from a black-box score to a trusted, actionable decision

Team Visionaries

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